📘 **1. What is the Objective of AI?**
Answer:
The objective of Artificial Intelligence (AI) is to develop **intelligent systems** that can perform tasks which normally require **human intelligence**.
🔹 **Key Objectives:**

- To **simulate human intelligence** in machines
- To enable machines to **learn, reason, and make decisions**
- To solve **complex problems efficiently**
- To automate tasks and improve **productivity**
- To provide **intelligent assistance** (e.g., virtual assistants)

🔹 **Example:**

- Chatbots like virtual assistants
- Self-driving cars

📘 **2. Define Artificial Intelligence (AI)**
🔹 **Answer:**
Artificial Intelligence (AI) is the branch of computer science that deals with the **creation of machines that can think, learn, and act like humans**.
🔹 **Key Points:**

- AI systems can **perceive environment**
- Can **learn from data**
- Can **make decisions**
- Works using algorithms and models

🔹 **Example:**

- Face recognition
- Recommendation systems

📘 **3. What is the Scope of AI?**
🔹 **Answer:**
The scope of AI refers to the **areas where AI techniques are applied** to solve real-world problems.
🔹 **Key Areas:**

- **Machine Learning**
- **Natural Language Processing (NLP)**
- **Robotics**
- **Computer Vision**
- **Expert Systems**

🔹 **Applications in Scope:**

- Healthcare
- Finance
- Education
- Transportation

🔹 **Key Point:**
👉 AI has a **wide and growing scope** in almost every industry

📘 **4. What are the Applications of AI?**
🔹 **Answer:**
AI is used in various real-world applications to improve efficiency and automation.
🔹 **Major Applications:**
🔸 **Healthcare**

- Disease prediction
- Medical diagnosis

🔸 **Finance**

- Fraud detection
- Algorithmic trading

🔸 **Education**

- Smart learning systems
- Personalized learning

🔸 **Transportation**

- Self-driving cars
- Traffic management

🔸 **Business**

- Chatbots
- Recommendation systems

🔸 **Robotics**

- Industrial automation

📘 🔹 **INTELLIGENT AGENT**
1. What is an Intelligent Agent? ⭐
Definition:
An Intelligent Agent is an entity that **perceives its environment through sensors and acts upon it using actuators** to achieve goals.
Key Points:

- Takes input (percepts) from environment
- Processes information
- Produces actions

- Works to achieve specific goals

Example:

- Robot, self-driving car, chatbot

2. Define Rational Agent
Definition:
A Rational Agent is an agent that **always chooses the best possible action** to maximize performance based on available information.
Key Points:

- Uses performance measure
- Acts logically and optimally
- Depends on percept history

Example:

- GPS navigation choosing shortest path

3. Types of Agents
Types:

1. **Simple Reflex Agent**
   - Works on condition-action rules
2. **Model-Based Agent**
   - Maintains internal state
3. **Goal-Based Agent**
   - Acts to achieve goals
4. **Utility-Based Agent**
   - Chooses best action based on utility
5. **Learning Agent**
   - Learns from experience

4. What is PEAS Description?
Definition:
PEAS stands for **Performance measure, Environment, Actuators, Sensors**.
Purpose:
Used to define the **task environment of an agent**
Example (Self-driving car):

- P → Safety, speed
- E → Roads, traffic
- A → Steering, brakes
- S → Cameras, sensors

📘 🔹 **PROBLEM SOLVING**
5. What is Problem Formulation in AI?
Definition:
Problem formulation is the process of **defining a problem in terms of initial state, goal state, and actions**.
Components:

- Initial state
- Actions (operators)
- Goal state
- Path cost

Example:

- 8-puzzle problem

6. Define State Space ⭐
Definition:
State space is the **set of all possible states** that can be reached from the initial state.
Key Points:

- Represented as graph/tree
- Nodes = states
- Edges = actions

Example:

- All possible configurations of a puzzle

7. What is Goal Test?
Definition:
Goal test is a method to check whether the **current state satisfies the goal condition**.
Key Point:

- Used to determine when to stop search

Example:

- Checking if puzzle is solved

8. Define Operators
Definition:
Operators are **actions that move the system from one state to another**.
Key Points:

- Change state
- Define transitions

Example:

- Move left/right/up/down in puzzle

📘 🔹 **UNINFORMED SEARCH**
1. What is Uninformed Search?
Definition:
Uninformed search (Blind search) is a search method that **does not use any domain knowledge or heuristic information**.
Key Points:

- Uses only problem definition

- Explores nodes systematically
- Examples: BFS, DFS

2. Define BFS (Breadth First Search) ⭐
Definition:
BFS is a search algorithm that **explores nodes level by level** starting from the root.
Key Points:

- Uses **Queue (FIFO)**
- Finds **shortest path** (optimal for equal cost)
- Complete

Example:
Level-wise traversal of a tree

3. Define DFS (Depth First Search) ⭐
Definition:
DFS is a search algorithm that **explores as deep as possible before backtracking**.
Key Points:

- Uses **Stack (LIFO)**
- Uses less memory
- May not be optimal

Example:
Explore one branch completely before moving to next

4. What is Iterative Deepening Search (IDS)?
Definition:
IDS is a combination of BFS and DFS that **repeatedly applies DFS with increasing depth limits**.
Key Points:

- Uses less memory
- Finds optimal solution
- Suitable for large search space

5. What is Bidirectional Search?
Definition:
Bidirectional search runs **two searches simultaneously**:

- One from start
- One from goal

Key Points:

- Meet in middle
- Faster than normal search
- Reduces search time

📘 🔹 **INFORMED SEARCH**

6. What is a Heuristic Function? ⭐ ⭐
Definition:
A heuristic function **h(n)** estimates the cost from current node to goal.
Key Points:

- Guides search efficiently
- Used in informed search
- Should be **admissible (not overestimate)**

Example:
Distance to destination in GPS

7. Define Greedy Search
Definition:
Greedy Best First Search selects node with **lowest heuristic value h(n)**.
Key Points:

- Focuses only on goal
- Fast but not always optimal

8. Define A* Algorithm ⭐ ⭐
Definition:
A* is an informed search algorithm that uses evaluation function:
👉 **f(n) = g(n) + h(n)**
Where:

- g(n) = cost from start
- h(n) = estimated cost to goal

Key Points:

- Finds optimal solution
- Uses both actual + heuristic cost
- Widely used in AI

9. What is Evaluation Function f(n)?
Definition:
Evaluation function f(n) determines the **best node to expand**.
👉 **Formula:**
f(n) = g(n) + h(n)
Key Points:

- Used in A* search
- Combines actual and estimated cost

Here are your **exam-ready answers (2–5 marks, high scoring format)** 📘 🔥
👉 Follow: **Definition + Key points + Example**
📘 🔹 **LOCAL SEARCH**

1. What is Hill Climbing? ⭐
Definition:
Hill Climbing is a **local search algorithm** that starts from an initial state and continuously moves to a **better neighboring state** based on heuristic value.
Key Points:

- Uses **heuristic function**
- Selects best immediate neighbor
- Stops when no better state exists
- Does not backtrack

Example:
Finding highest peak in a landscape

2. Limitations of Hill Climbing
Key Problems:

- **Local Optimum:** Gets stuck at local best solution
- **Plateau:** No improvement in neighbors
- **Ridge Problem:** Difficult to move in right direction
- Not guaranteed to find global optimum

3. What is Local Optimum?
Definition:
A local optimum is a state that is **better than its neighbors but not the best overall (global optimum)**.
Example:
Climbing a small hill instead of the highest mountain
📘 🔹 **ADVANCED TOPICS**
4. What is AO* Algorithm?
Definition:
AO* is an **informed search algorithm** used for solving problems represented as **AND-OR graphs**.
Key Points:

- Expands most promising node
- Considers both AND and OR nodes
- Finds optimal solution path

5. Define Constraint Satisfaction Problem (CSP) ⭐
Definition:
A CSP is a problem where we must **assign values to variables** such that all **constraints are satisfied**.
Components:

- Variables
- Domains
- Constraints

6. Examples of CSP

- Sudoku
- N-Queens problem
- Map coloring problem

📘 🔴 **UNIT 3: GAME PLAYING**
7. What is Game Playing in AI?
Definition:
Game playing in AI involves designing algorithms that enable computers to **play games intelligently**.
Key Points:

- Used in two-player games
- Uses search techniques
- Examples: Chess, Tic-Tac-Toe

8. Define Minimax Algorithm ⭐ ⭐
Definition:
Minimax is a decision-making algorithm used in **two-player games** where one player tries to maximize score and the other tries to minimize it.
Key Points:

- MAX player → maximize value
- MIN player → minimize value
- Works on game tree
- Used for optimal decisions

9. What is Alpha-Beta Pruning? ⭐ ⭐
Definition:
Alpha-Beta pruning is an optimization technique of Minimax that **eliminates unnecessary branches** in the game tree.
Key Points:

- Reduces computation
- Uses α (alpha) and β (beta) values
- Prunes branches when $\alpha \geq \beta$

10. What is Game Tree?
Definition:
A game tree is a **tree representation of all possible moves** in a game.
Key Points:

- Nodes → game states
- Edges → moves
- Used in Minimax

11. What is Utility Function?
Definition:
Utility function assigns a **numerical value to a game state** representing how good or bad it is.
Key Points:

- Used in decision making
- Helps choose best move

Example:
Win = +1, Loss = -1, Draw = 0

Here are your **exam-ready answers (2–5 marks, high scoring format)** 📘 🔥
👉 Use: **Definition + Key points + Example (if possible)**

1. Define Propositional Logic ⭐
Definition:
Propositional Logic is a formal system where statements (propositions) are either **true or false**.
Key Points:

- Uses logical connectives: AND (∧), OR (∨), NOT (¬)
- Does not deal with variables or quantifiers
- Simple representation

Example:
P: "It is raining"
Q: "It is cold"
P ∧ Q

2. What is Predicate Logic (FOL)? ⭐ ⭐
Definition:
First Order Logic (FOL) extends propositional logic by including **predicates, variables, and quantifiers**.
Key Points:

- Represents relationships between objects
- Uses quantifiers:
    - ∀ (for all)
    - ∃ (there exists)
- More expressive than PL

Example:
∀x Human(x) → Mortal(x)

3. Difference between PL and FOL ⭐

Basis	Propositional Logic	Predicate Logic
Representation	Simple statements	Uses variables & predicates
Expressiveness	Limited	More powerful
Quantifiers	Not used	Used (∀, ∃)
Example	P ∧ Q	∀x P(x)

4. What is a Knowledge Base?
Definition:
A Knowledge Base is a collection of **facts and rules** used by an AI system to make decisions.
Key Points:

- Stores information
- Used for reasoning
- Works with inference engine

5. What is Inference? ⭐
Definition:
Inference is the process of **deriving new knowledge** from existing facts and rules.
Key Points:

- Uses logical reasoning
- Types: Forward chaining, Backward chaining

6. What is Resolution? ⭐ ⭐
Definition:
Resolution is a **rule of inference** used to prove statements by converting them into **CNF (Conjunctive Normal Form)**.
Key Points:

- Used in theorem proving
- Eliminates contradictions
- Works on clauses

7. What is Planning in AI?
Definition:
Planning is the process of **finding a sequence of actions** to achieve a goal.
Key Points:

- Starts from initial state
- Ends at goal state
- Used in robotics, scheduling

8. Define Partial Order Planning
Definition:
Partial Order Planning is a planning technique where actions are **not strictly ordered**, allowing flexibility.
Key Points:

- Actions executed when required
- Reduces unnecessary constraints

9. What is Uncertain Knowledge?
Definition:
Uncertain knowledge refers to situations where information is **incomplete, ambiguous, or probabilistic**.
Key Points:

- Cannot guarantee exact outcome
- Uses probability for reasoning

10. Define Probability in AI
Definition:
Probability in AI is used to **measure uncertainty** and predict outcomes.

Formula:
P(A) = Probability of event A
Key Points:

- Range: 0 to 1
- Used in decision making

11. What is a Bayesian Network? ⭐ ⭐
Definition:
A Bayesian Network is a **graphical model** that represents probabilistic relationships among variables.
Key Points:

- Uses Directed Acyclic Graph (DAG)
- Nodes = variables
- Edges = dependencies
- Uses conditional probability

Example:
Rain → Wet Grass
Here are your **exam-ready answers (2–5 marks, high scoring format)** 📘 🔥
👉 Use: **Definition + Key points + Example**
🔹 **1. What is Machine Learning?** ⭐
Definition:
Machine Learning is a branch of AI that enables systems to **learn from data and improve performance without explicit programming**.
Key Points:

- Learns from experience
- Uses algorithms and models
- Improves automatically

Example:
Spam email detection
🔹 **2. Define Supervised Learning** ⭐ ⭐
Definition:
Supervised Learning is a type of ML where the model is trained using **labeled data**.
Key Points:

- Input + Output given
- Learns mapping function
- Used for classification & regression

Example:
Predicting house prices
🔹 **3. Define Unsupervised Learning** ⭐ ⭐
Definition:
Unsupervised Learning is a type of ML where the model works on **unlabeled data**.
Key Points:

- Finds hidden patterns
- No predefined output
- Used for clustering

Example:
Customer segmentation
🔹 **4. Difference: Supervised vs Unsupervised** ⭐

Basis	Supervised	Unsupervised
Data	Labeled	Unlabeled
Output	Known	Unknown
Task	Prediction	Pattern finding
Example	Classification	Clustering

5. What is a Decision Tree? ⭐
Definition:
A Decision Tree is a model that uses a **tree structure to make decisions**.
Key Points:

- Nodes → decisions
- Branches → conditions
- Leaves → outcomes

Example:
Play Tennis decision problem
6. What is SVM (Support Vector Machine)?
Definition:
SVM is a supervised learning algorithm that **finds the optimal boundary (hyperplane)** to separate data.
Key Points:

- Used for classification
- Maximizes margin
- Works with high-dimensional data

7. What is a Neural Network? ⭐
Definition:
A Neural Network is a system inspired by the human brain that consists of **interconnected nodes (neurons)**.
Key Points:

- Input, hidden, output layers
- Learns patterns
- Used in deep learning

Example:
Image recognition
8. What is Market Basket Analysis?
Definition:
Market Basket Analysis is a technique used to **identify relationships between items** purchased together.
Key Points:

- Uses association rules
- Helps in recommendation systems

Example:
People buying bread also buy butter

🔹 **9. What is Natural Language Processing (NLP)?** ⭐

Definition:
NLP is a field of AI that enables machines to **understand and process human language**.

Key Points:

- Speech recognition
- Text processing
- Language translation

Example:
ChatGPT, Google Translate

🔹 **10. What are Issues in NLP?** ⭐

Key Issues:

- **Ambiguity** (multiple meanings)
- **Context understanding**
- **Syntax & semantics**
- **Language diversity**

🔹 **11. What is an Expert System?** ⭐

Definition:
An Expert System is an AI system that **emulates the decision-making ability of a human expert**.

Key Points:

- Uses knowledge base
- Uses inference engine
- Solves complex problems

🔹 **12. Components of Expert System** ⭐

- **Knowledge Base** (facts + rules)
- **Inference Engine** (reasoning)
- **User Interface**
- **Explanation System**

🔹 **13. What is Robotics in AI?**

Definition:
Robotics in AI deals with designing machines (robots) that can **perform tasks automatically using intelligence**.

Key Points:

- Uses sensors & actuators
- Works in real-world environments
- Combines AI + hardware

Example:
Industrial robots, self-driving cars